

Mortality age

A reproducible, age-free, sex-dependent, multi-cohort mortality-equivalent age experiment

Version 0.2 · 9 October 2026 · Longevity World Cup

Revision: Chronological age is absent from every measurement-to-risk equation and every missing-measurement regression. All models and sensitivities have been refit. The calculator requires sex and measurements; it neither requests birth date nor substitutes a default or inferred personal age. A separate, fixed historical age reference converts risk into years.

Age-free scoring does not establish age-independent measurement: NHANES treadmill testing uses age-informed protocols and an age-predicted maximal heart rate. This revision retains its released VO2 estimate; upstream fitness-method age dependence remains a limitation. [5-7]

Finding: NHANES supports several jointly measured mortality models, but these public cohorts do not establish a validated clock using all eight requested domains. No respondent has the complete panel. The complete-panel equation remains an unvalidated integration experiment. Previously inspected temporal cohorts are reused for exploratory diagnostics.

The development integration includes 12,645 people aged 18-49 and 475 deaths over available follow-up. Marginal temporal evaluation includes 8,182 people, 183 total deaths, and 78 deaths within five years. Missing measurements are integrated out during this evaluation; this is **not validation of a jointly observed eight-domain prediction**.

In marginal temporal evaluation, the full experiment has observed/expected five-year mortality 1.08 and unweighted Harrell C 0.659. The observed inflammation/kidney panel has observed/expected 1.05, slope 1.00 and C 0.795. Its broader age range prevents interpreting this as a head-to-head comparison.

Conditional paired survey-bootstrap error comparisons are age/sex: includes zero; age-free common core: favors full integration. These are marginal comparisons in reused cohorts; even a positive result cannot establish complete-panel validity.

The fitness panel has only 4 five-year deaths in its temporal test, with observed/expected 0.19. Its numeric panel age is withheld in the calculator. Other panel estimates and any full integration result remain experimental, not clinical validation.

The displayed age matches modeled five-year all-cause mortality to a historical sex-specific age reference. It measures an association in selected survey populations. It does not estimate pace of aging, life expectancy, treatment benefit, or years gained.

Calculator: longevityworldcup.com/mortality-age

Frozen model: [mortality-age-model.json](#)

1. Measurements and populations

Domain	Definition and canonical unit
Circulation	Mean nonzero readings 1-3 of resting clinic systolic and diastolic BP, mmHg. Both enter the joint equation. Not 24-hour blood pressure. [4]
Fitness	CVDESVO2, exercise-based treadmill estimate, mL/kg/min. Do not substitute CVDVOMAX, the non-exercise protocol-assignment estimate. Eligibility excluded many people with health conditions; ages 12-49 were tested. Adult research population: 18-49; removing the age input does not validate use in older people. [5-7]
Strength	Highest MGX trial from either hand with effort code 1 (maximal). Unit kg. Not MGDCGSZ, which adds best readings from both hands. [8-9]
Body composition	Measured waist cm / standing height cm. The same unit is required in numerator and denominator. [10]
Lipids	LBXAPB ApoB, mg/dL, from the morning fasting subsample. [11-12]
Metabolism	LBXGH HbA1c, NGSP percent. IFCC conversion: % = $0.09148 \times \text{mmol/mol} + 2.152$. [13]
Kidney	SSCYST cystatin C, mg/L, revised SSCARD_A stored-sample release spanning 1999-2004. Laboratory testing was in 2018-2020. Dedicated updated sample weights are used for its distributions and observed panels. [3]
Inflammation	CRP: early LBXCRP mg/dL $\times 10$; 2015 LBXHSCRIP is already mg/L. Released below-detection fill values are retained. The 2015 assay is separate sensitivity data. [14-15]

Recorded RIAGENDR categories (female/male) are required. Every component includes a regularized sex-dependent slope; sex is also included in the baseline and age reference. These categories do not encode gender identity or hormone treatment. Differences that shrink toward zero are uncertain, not proof that sexes have identical effects.

NHANES uses age in treadmill protocol assignment and the predicted maximal heart rate (220 minus age). This revision removes personal age from scoring and missing-measurement conditioning; it does not redesign or remove age from upstream fitness estimation. Strictly age-independent fitness measurement would require a separately validated method and calibration. [5-7]

Analytic rows are mortality-eligible, nonpregnant where identified, aged 18-79, with event status 0/1 and positive examination follow-up. Ages 80+ are excluded because of top-coding. Unknown pregnancy is not equivalent to a measured negative pregnancy test. The common core requires BP, waist/height and HbA1c. Complete-core selection and measurement exclusions may affect transport.

2. Data, overlap and frozen split

Public NHANES 1999-2016 is linked by SEQN to the 2019 public-use mortality files, using PERMTH_EXM and MORTSTAT. All causes are included. Public follow-up can be perturbed for confidentiality; mortality status is preserved. Restricted newer linkage was not accessed. [1-2]

Development cycles: 1999, 2001, 2005, 2007, 2011. Temporal diagnostics: 2003, 2009, 2013. The 2015 cycle supplies a separate two-year assay/short-follow-up sensitivity. A deterministic respondent hash separates internal 80% development fitting from 20% model selection. The original analysis-plan.md preceded version 0.1 fitting; age-free-plan.md was saved before version 0.2 refitting.

Version 0.1 review corrected combined-cycle scaling and latent residual subsample weights. Version 0.2 removes chronological age from risk and missing-input distributions, refitting with the same cycles, weights, internal candidate grid and seeds. All these temporal cohorts/endpoints were previously inspected. This is exploratory diagnostic reuse, not fresh independent validation; no model is retuned from these diagnostics.

Cycle	Core N / D	CRP+cys N / D	ApoB+CRP N / D	ApoB+grip N / D	CRP+cys+VO2 N / D
1999-00	3,944 / 976	3,502 / 878	0 / 0	0 / 0	1,137 / 49
2001-02	4,257 / 810	3,965 / 760	0 / 0	0 / 0	1,379 / 42
2003-04	4,006 / 726	3,762 / 682	0 / 0	0 / 0	1,325 / 25
2005-06	4,068 / 523	0 / 0	1,935 / 246	0 / 0	0 / 0
2007-08	4,848 / 606	0 / 0	2,346 / 308	0 / 0	0 / 0
2009-10	5,191 / 429	0 / 0	2,503 / 202	0 / 0	0 / 0
2011-12	4,578 / 302	0 / 0	0 / 0	2,045 / 143	0 / 0
2013-14	4,955 / 216	0 / 0	0 / 0	2,234 / 80	0 / 0
2015-16	4,716 / 130	0 / 0	2,085 / 56	0 / 0	0 / 0

N = observed complete rows for that panel; D = deaths over all available follow-up, not five years. Panels include the common core. These availability counts precede positive-weight exclusions. **All-eight complete cases: zero in every cycle.** Exercise fitness and grip were never collected together in these cycles.

Model	Development N / D	Temporal N / D / D≤5y
Common core	21,695 / 3217	14,152 / 1371 / 495
Core + CRP + cystatin	7,467 / 1638	3,762 / 682 / 156
Core + ApoB + CRP	4,281 / 554	2,503 / 202 / 79
Core + ApoB + grip	2,045 / 143	2,234 / 80 / 64
Core + CRP + cystatin + fitness	2,516 / 91	1,325 / 25 / 4

The age/sex reference uses the complete-core development population aged 18-79, not every eligible NHANES respondent. Modelled five-year mortality is therefore a historical complete-core reference, not a forecast of current national mortality.

3. Survival model and sex-dependent components

Follow-up t is in years; d is death status. Sex s is 0 for female and 1 for male. For observed inputs, the sex-specific Gompertz model is:

$$\begin{aligned}\eta(x,s) &= \alpha + \delta s + \sum_j f_j(x_j, s) \\ h(t|x,s) &= \exp(\eta) \exp(\gamma_s t) \\ H(t|x,s) &= \exp(\eta) A(\gamma_s, t), \text{ where } A(\gamma, t) = (\exp(\gamma t) - 1)/\gamma \\ P(\text{death by 5 years}) &= 1 - \exp[-H(5)]\end{aligned}$$

Chronological age, birth date, age group and inferred personal age are not predictors. Research age determines cohort eligibility and diagnostic subgroup comparisons only. Thus biomarker associations can include differences related to chronological age; removing the age predictor does not establish a causal or uniquely biological-aging effect.

Log transformations are used for ApoB, CRP, cystatin C, grip and fitness to model positive measurements. Other measurements use their raw scale. Development-weighted 10th/50th/90th percentiles are the three restricted-cubic-spline knots. z is centered/scaled transformed input; q is centered/scaled spline basis. The component is:

$$f_j(x,s) = \beta_j z_j(x) + \theta_j q_j(x) + \psi_j z_j(x)(s - 0.5)$$

This is $f(\text{VO2max}, \text{sex})$, $f(\text{grip}, \text{sex})$, and equivalent sex-dependent functions for the other markers. Shared nonlinear curvature plus sex-specific slope deviations borrow information. Separate unrestricted male/female nonlinear fits would consume more information than the young-event counts support. No biomarker is capped to a preferred clinical value, and U/J shapes are not forced.

Three knots $k_1 < k_2 < k_3$ define q before centering/scaling: $[(u-k_1)_+^3 - (u-k_2)_+^3(k_3-k_1)/(k_3-k_2) + (u-k_3)_+^3(k_2-k_1)/(k_3-k_2)]/(k_3-k_1)^2$. Its tails are linear mathematically; the calculator refuses values beyond declared empirical support instead of extrapolating those tails.

The weighted negative mean log-likelihood uses ridge shrinkage. Relative penalties are 1 for main linear terms, 4 for nonlinear terms, 8 for sex deviations, 0.05 for sex and zero for intercept. The internal grid is linear/log $\lambda = 0.0003, 0.001, 0.003$ and smooth $\lambda = 0.001, 0.003$. The selected smooth model with $\lambda = 0.001$ wins by internal marginal survival loss and is refit on development only.

Candidate	Internal loss	Converged
Linear/log, $\lambda=0.0003$	0.220108	True
Linear/log, $\lambda=0.001$	0.219529	True
Linear/log, $\lambda=0.003$	0.219239	True
Smooth, $\lambda=0.001$	0.218460	True
Smooth, $\lambda=0.003$	0.218483	True

Biomarker coefficients are unconstrained in sign. Positive age slopes exist only in the separate historical reference, for inversion. Follow-up slopes are bounded -0.1 to 0.2 per year. These are optimizer constraints, not biomarker clipping. Survey weights are normalized by sum in the likelihood. The fit includes full right-censored follow-up; its five-year output depends on the time form and proportional-effects assumption.

4. What combining cohorts assumes

Adding adjusted marginal hazard ratios would double count shared information and would not recover a joint model. Here each original participant contributes a survival likelihood marginalized over their unobserved measurements:

$$L_i = E_{X_{\text{missing}} | X_{\text{observed}}, \text{sex}, \text{core}} \{ h(t_i | X, s)^{d_i} \exp[-H(t_i | X, s)] \}$$

Conditional measurement means are ridge regressions on sex, the four common-core measurements and their sex interactions: ten predictors including the intercept. Age, age² and age×sex have been removed. Residuals for the five occasionally measured inputs are approximated by sex-specific multivariate Gaussian distributions on standardized transformed scales. Observed values remain fixed during integration. The draws are quadrature, not fabricated observed participants.

Genuinely observed residual pair covariance is retained. A pair with fewer than 50 jointly measured records in a sex is assigned zero residual covariance in the primary model. Unobserved relationships cannot be inferred merely because datasets share the common core. The zero-residual-dependence choice and the transport of conditional distributions/outcome relationships across cycles and fitness exclusions are assumptions. [16-17]

Pair	Female N	Male N
apob + crp	1147	1280
apob + cystatin	0	0
apob + grip	547	632
apob + vo2	0	0
crp + cystatin	2158	2309
crp + grip	0	0
crp + vo2	1293	1437
cystatin + grip	0	0
cystatin + vo2	1193	1323
grip + vo2	0	0

Conditional means and residual covariances use dedicated ApoB fasting weights or revised cystatin weights for the involved subsample markers, otherwise MEC weights. Observed-panel survival uses the panel-specific weights. Per CDC guidance, early four-year weight numerators are twice the two-year numerators; a common normalization cancels. The full outcome model uses MEC weights; no published joint subsample weight covers all eight domains. Fitness noncompletion lacks an official fitness response weight. These limitations affect population calibration. [3,18,21]

The primary integral uses 32 deterministic antithetic Gaussian draws; 64-draw refitting checks numerical sensitivity. Residual covariance matrices are shrunk toward their diagonal only if needed for positive definiteness. Unknown pair correlations are changed to -0.3, +0.3 and +0.6, then refit. PSD shrinkage changes the attained correlations; all matrices and shrinkage are saved.

5. Temporal evaluation and its limits

Model	O/E (80% PSU range)	C	Slope	Brier 5y
Full integration (marginal)	1.08 (0.95 to 1.22)	0.659	0.82	0.00931
Common core	0.97 (0.91 to 1.03)	0.704	0.78	0.02721
Core + CRP + cystatin	1.05 (0.95 to 1.13)	0.795	1.00	0.02500
Core + ApoB + CRP	0.90 (0.79 to 1.00)	0.719	0.85	0.02502
Core + ApoB + grip	0.88 (0.75 to 1.01)	0.746	0.68	0.02320
Core + CRP + cystatin + fitness	0.19 (0.09 to 0.31)	0.598	Not estimable	0.00124

O/E = weighted Kaplan-Meier observed five-year death probability / weighted mean predicted probability; 1 is the target. Its 80% interval uses 128 survey-PSU bootstraps within strata with the model fixed. C is unweighted Harrell concordance over available censored follow-up, not five-year AUC. Slopes use a five-year truncated Cox recalibration on log cumulative risk when at least 30 five-year events are available. A missing slope is insufficient event information. Brier error uses inverse-censoring-probability weights with the censoring distribution estimated in the evaluation cohort. Section 7.2 gives conditional survey-bootstrap ranges for C, slope and Brier error.

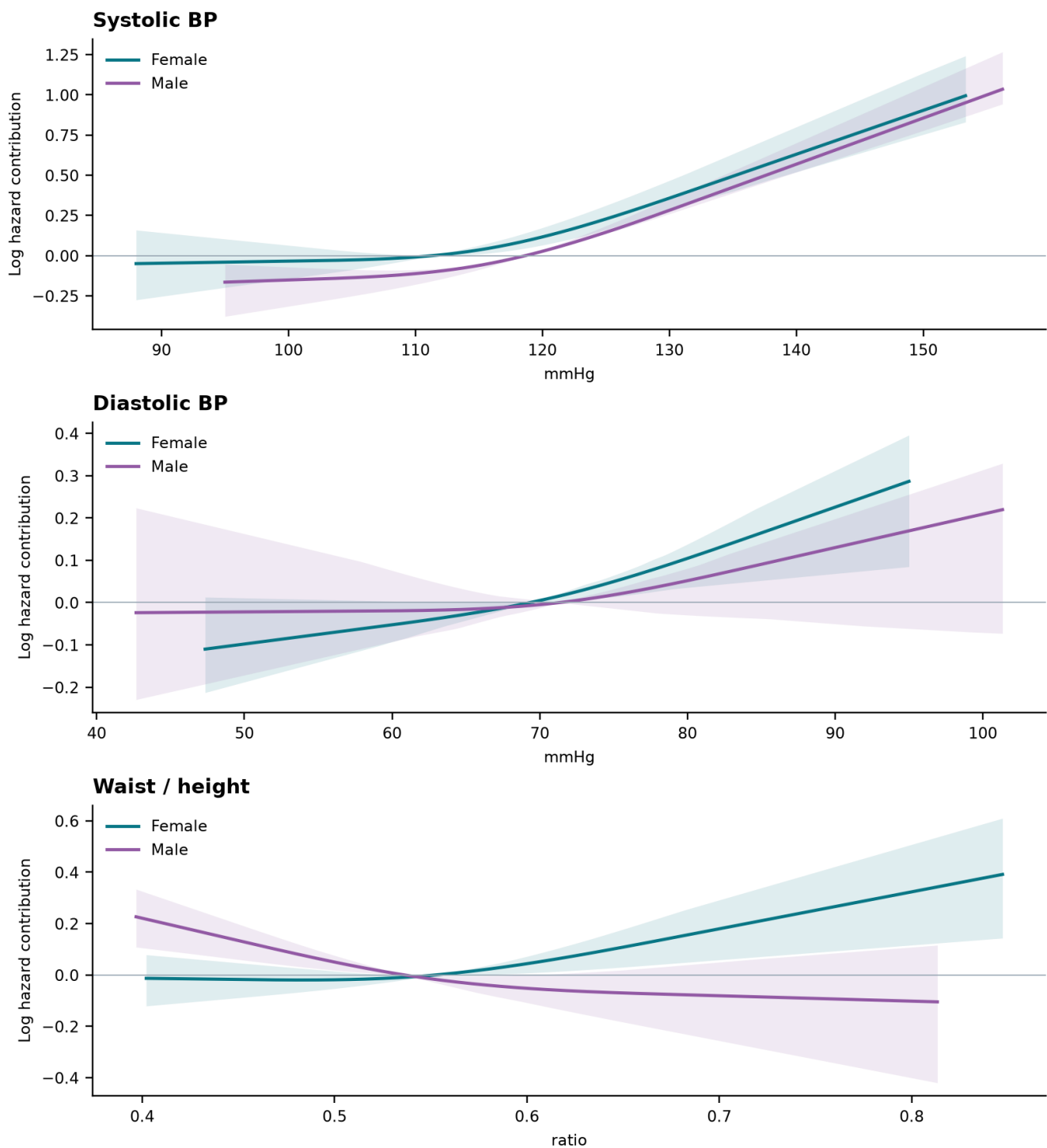
Same under-50 holdout	O/E	C	Brier 5y
Age/sex	1.45	0.649	0.009325
Observed common core	0.46	0.579	0.009576
Full marginal integration	1.08	0.659	0.009310

The full marginal Brier improvement over the age/sex research benchmark is +0.000015; positive favors integration. Personal age enters the benchmark only, not the clock prediction. These predictions integrate missing measurements. Even good marginal calibration would not prove that the unseen joint eight-domain equation is correct. There is no external cohort validation.

Full marginal test	N	Deaths ≤5y	O/E (80% PSU range)
Female	4102	26	0.90 (0.73 to 1.08)
Male	4080	52	1.19 (1.00 to 1.36)

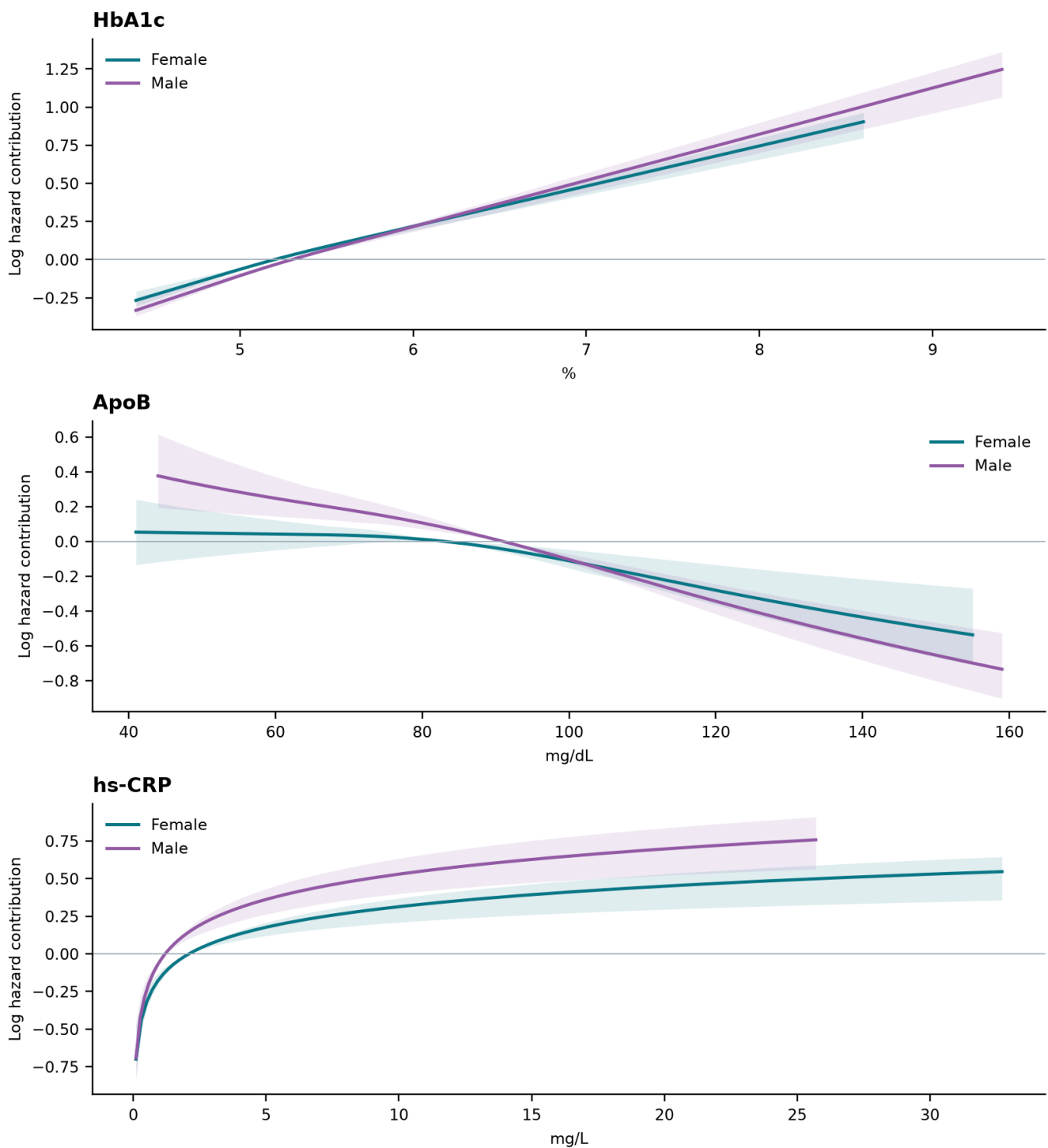
The fitness panel has only 4 five-year deaths and O/E 0.19; it cannot supply reliable calibration or elaborate sex-specific fitness curves. The version 0.1 decision to withhold its numeric age is retained. The frozen model and all diagnostic metrics remain available. Partial-panel results do not establish a validated full clock.

6.1 Sex-dependent curves and conditional uncertainty



Each curve is the log-hazard contribution relative to that sex's development median profile, holding other inputs fixed. Shading is the 10th-90th percentile of 32 survey-cluster refits with fixed knots, conditional distributions, selected specification and age reference. It omits nuisance-estimation and unobserved-dependence uncertainty. Limits are development-weighted 1st-99th input percentiles by sex, not clinical normal ranges.

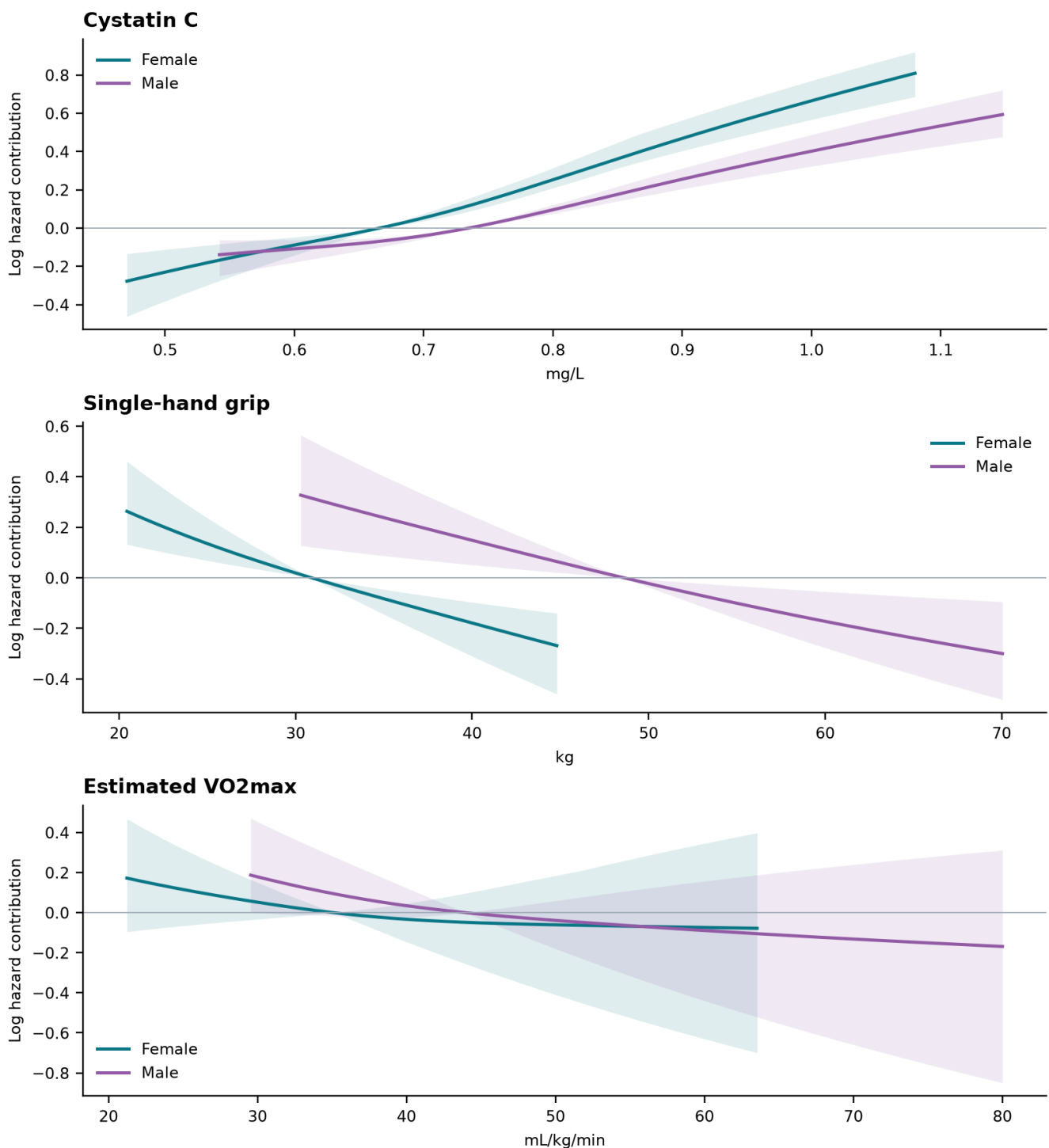
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These are jointly fitted age-free associations. Inverse, flat or nonmonotone portions cannot be interpreted as benefits from changing a marker. Selection, illness, chronological-age differences, confounding and integration assumptions can affect the fitted shape.

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Sparse outcomes and selective exercise eligibility limit interpretation of fitness and strength. Shrunk sex deviations do not establish equivalence between sexes. The age-free refit can change each curve because age is no longer adjusted out.

7. Sensitivity, uncertainty and failed hypotheses

Refit / assumption	Female example age	Male example age
Primary 32-draw integration	37.93	34.91
Unknown residual correlation -0.3	37.74	35.06
Unknown residual correlation +0.3	37.52	34.13
Unknown residual correlation +0.6	37.28	34.11
64-draw integration	37.80	34.76
Exponential baseline + matched reference	34.68	32.10

Examples use sex-specific development medians and contain no chronological age. They are artificial profiles, not two jointly observed people. The exponential sensitivity refits both the age-free outcome model and historical age reference with constant baseline hazards. It demonstrates time-form dependence; it does not establish the proportional-effects assumption.

Exploratory sex-stratified Cox diagnostics using the observed core/blood spline bases found ranked-time associations in scaled Schoenfeld residuals. The smallest approximate p values are core: hba1c_sex, $p=3.6e-08$; blood: sbp_sex, $p=0.0028$. This raises concern about constant covariate effects over follow-up. These penalized, survey-weighted diagnostics have no survey-design or multiple-testing correction and are not a formal test of the integrated likelihood.

One-at-a-time sweeps across sex and nine supported values per input produced 162 age-free artificial profiles. The median tested dependence age span is 1.11 years; the maximum is 6.33 years. 0 primary profiles lie outside the reference-age support. This is not a simultaneous joint-tail stress test and does not exhaust unidentified relationships.

Two-year landmark fits restart follow-up among people surviving past two years and subtract 24 months. No personal age enters these refits or missing-measurement draws. Baseline measurements remain two years old. The corrected time origin does not remove survivor selection or reverse causality. Integrated and observed-panel landmark artifacts are exploratory sensitivities.

The 32 training bootstraps are conditional on fixed nuisance distributions, knots, model selection and reference; their 80% age/curve ranges are not comprehensive prediction intervals. Structural sensitivity spans are separate, finite scenario comparisons, not confidence intervals. Regularization may conceal weakly identified components. One must not infer that each domain is independently measured biological aging.

The proposed all-domain clock hypothesis remains unestablished. Additional joint measurements, mortality events, sex-specific evaluation, measurement-method calibration and independent cohort validation are required before clinical or intervention interpretation. Accessible public NHANES overlaps cannot supply the absent exercise/grip joint information. HRS health access and UK Biobank would require separately authorized access; neither dataset was used. [19-20]

7.1 Later assays and longer follow-up

The two-year later-assay and ten-year temporal diagnostics were introduced by transport-plan.md in version 0.1. Version 0.2 recomputes these previously inspected endpoints under age-free-plan.md; this is diagnostic reuse. Horizons follow administrative coverage. Predictors, measurement distributions and release decisions remain fixed during evaluation. These are not new confirmatory validation.

2015-2016: newer hs-CRP assay, two-year endpoint

Model	N	D≤2y	O/E (80% PSU range)	Slope	Brier 2y
Full integration (marginal)	2,604	8	1.08 (0.65 to 1.46)	Not estimable	0.003634
Common core	4,716	68	1.01 (0.81 to 1.20)	0.90	0.011567
Core + ApoB + CRP	2,085	31	0.85 (0.64 to 1.04)	1.01	0.010272

2003-2004: ten-year endpoint

Model	N	D≤10y	O/E (80% PSU range)	Slope	Brier 10y
Full integration (marginal)	2,381	44	0.78 (0.66 to 0.90)	0.75	0.015678
Common core	4,006	373	0.96 (0.92 to 1.01)	1.00	0.058505
Core + CRP + cystatin	3,762	349	1.05 (0.99 to 1.10)	1.06	0.054291
Core + CRP + cystatin + fitness	1,325	9	0.34 (0.24 to 0.44)	Not estimable	0.005044

The later cycle has at least 2.83 years of observed administrative follow-up, so it supports the two-year check and is not presented as five-year validation. Its population, calendar period and selection also differ; the comparison cannot isolate the assay effect. The 2003 cycle has at least 14.75 years of administrative follow-up. Its ten-year results reuse an already inspected temporal cohort.

Same-row full marginal comparison	Age/sex Brier	Core Brier	Full Brier
2015, 2 years	0.003620	0.003687	0.003634
2003, 10 years	0.015317	0.017630	0.015678

The table compares frozen predictions on identical rows and weights. The later under-50 cohort has 8 deaths by two years. These sparse, reused endpoints cannot establish unseen joint-panel validity, regardless of the direction of the point-error comparisons. Fitness remains numerically withheld.

Full marginal sex coverage	Female N / D≤h	Male N / D≤h
2015, 2 years	1,328 / 2	1,276 / 6
2003, 10 years	1,132 / 15	1,249 / 29

Ranges use 128 conditional PSU bootstraps with fixed models. Sparse sex strata cannot establish calibration. Exact sex-specific calibration, concordance over available follow-up, follow-up counts and same-row benchmarks are in the saved artifacts. The new evaluator agrees with the original five-year evaluator and an analytical censoring example. Diagnostic plan SHA-256: 4c6643498cdeab3fbff66da04a2e0ded67cff3e887d8d868fc979a07d4ac304f

7.2 Validation uncertainty and paired errors

These ranges quantify evaluation-sample uncertainty conditional on the frozen predictors, preprocessing, missing-input distributions, model selection and reference. Version 0.1 introduced uncertainty-plan.md; the age-free revision repeats the diagnostics under age-free-plan.md in the same previously inspected cohorts. No predictor is refit or chosen from these results. The intervals do not resolve training or transport uncertainty.

Five-year model	C (80% range)	Slope (80% range)	Brier (80% range)
Full integration (marginal)	0.659 (0.638 to 0.679)	0.82 (0.66 to 0.96)	0.009310 (0.008174 to 0.010458)
Common core	0.704 (0.697 to 0.712)	0.78 (0.72 to 0.84)	0.027206 (0.025916 to 0.028634)
Core + CRP + cystatin	0.795 (0.789 to 0.801)	1.00 (0.94 to 1.06)	0.024996 (0.022654 to 0.026839)
Core + ApoB + CRP	0.719 (0.706 to 0.732)	0.85 (0.76 to 0.93)	0.025021 (0.021793 to 0.027411)
Core + ApoB + grip	0.746 (0.725 to 0.766)	0.68 (0.59 to 0.77)	0.023195 (0.019872 to 0.026446)
Core + CRP + cystatin + fitness	0.598 (0.562 to 0.650)	Not estimable (Unavailable)	0.001240 (0.000630 to 0.001966)

The same survey PSU-within-stratum resampling supplies 128 replicates. Weighted Brier/recalibration estimates use the original survey weights multiplied by PSU multiplicities. Unweighted concordance repeats whole sampled PSU rows with their integer multiplicities. Repeated rows are bootstrap copies, not newly observed people. C remains a measure over available follow-up, not a five-year AUC.

An interval requires at least 80% finite replicates. Counts are Full integration (marginal): brier 128/128, concordance 128/128, slope 128/128; Common core: brier 128/128, concordance 128/128, slope 128/128; Core + CRP + cystatin: brier 128/128, concordance 128/128, slope 128/128; Core + ApoB + CRP: brier 128/128, concordance 128/128, slope 128/128; Core + ApoB + grip: brier 128/128, concordance 128/128, slope 127/128; Core + CRP + cystatin + fitness: brier 128/128, concordance 128/128, slope 0/128. Slopes are unavailable below 30 five-year deaths or when too few replicates are estimable. The artifacts retain every estimable count; unavailable estimates are not invented.

Full marginal comparison	Brier improvement	80% paired PSU range
Versus age/sex	+0.000015	-0.000030 to +0.000067
Versus observed common core	+0.000266	+0.000179 to +0.000340

Improvement is benchmark Brier error minus full-model Brier error on the same under-50 rows, weights and paired PSU resamples. Positive values favor full integration. Whether a range spans zero is shown explicitly. These are conditional marginal diagnostics in reused cohorts and cannot establish the unobserved full joint prediction.

Point metrics agree with the original five-year evaluator; the paired-error implementation also passes an analytical censoring calculation. Additional diagnostic plan SHA-256: 4c6643498cdeab3fbff66da04a2e0ded67cff3e887d8d868fc979a07d4ac304f

7.3 Research age-stratum diagnostics

Chronological age is used here only to audit the recorded research population. Predictions and missing-measurement draws were calculated before forming these groups, without age. These reused temporal subgroups do not change the fitted equation or certify individual support.

Age-free model	Recorded age	N	D≤5y	O/E	C
Full integration (marginal)	18-39	5,639	39	0.98	0.659
Full integration (marginal)	40-49	2,543	39	1.23	0.616
Common core	18-39	5,639	39	0.39	0.597
Common core	40-49	2,543	39	0.56	0.524
Common core	50-59	2,164	82	1.08	0.580
Common core	60-79	3,806	335	1.52	0.554
Core + CRP + cystatin	18-39	1,594	7	0.29	0.589
Core + CRP + cystatin	40-49	618	11	0.55	0.623
Core + CRP + cystatin	50-59	472	20	1.23	0.686
Core + CRP + cystatin	60-79	1,078	118	1.51	0.628
Core + ApoB + CRP	18-39	940	4	0.34	0.565
Core + ApoB + CRP	40-49	480	6	0.63	0.626
Core + ApoB + CRP	50-59	414	12	0.59	0.643
Core + ApoB + CRP	60-79	669	57	1.67	0.580
Core + ApoB + grip	18-39	872	5	0.27	0.529
Core + ApoB + grip	40-49	399	7	1.16	0.775
Core + ApoB + grip	50-59	390	15	1.39	0.622
Core + ApoB + grip	60-79	573	37	0.93	0.678
Core + CRP + cystatin + fitness	18-39	1,073	3	0.17	0.552
Core + CRP + cystatin + fitness	40-49	252	1	0.24	0.637

O/E is observed/expected five-year risk; C is unweighted concordance over available follow-up. Sparse groups can show unstable ratios or unavailable discrimination. This table reports point diagnostics, not formal subgroup validation. Full and fitness data have no 50+ research participants; removing age from the calculator does not supply that evidence.

8. From risk to age, and supported inputs

Reference risk is fit in the complete-core development population aged 18-79. The same five-year endpoint and sex are used on both sides. Matching cumulative hazard rather than rounded probabilities gives a stable closed-form age:

$$\text{Age}_{\text{eq}} = 45 + 10 \{ \log H_{\text{model}}(5) - \log A(\gamma_{\text{ref},s}, 5) - \alpha_{\text{ref}} - \delta_{\text{ref},s} \} / b_{\text{ref},\text{age},s}$$

Reference parameter	Value
intercept	-6.75843283148
male	0.85644335816
age_female	0.920587062491
age_male	0.832440010619
gamma_0	0.120357788382
gamma_1	0.0921579396873

The reference is monotone because its age coefficients are positive. It is a fixed risk-to-years ruler: the input risk and sex determine equivalent age; the user's chronological age is never used. Inversion is allowed only over reference ages 18-79. The full/fitness research population is 18-49 and other panels are 18-79. The age-free calculator cannot enforce a personal age gate, and does not validate wider population use. Inputs outside sex-specific weighted 1st-99th development percentiles return unsupported rather than capped measurements or clipped output. Marginal ranges do not certify every joint combination.

The calculator withholds a sampling or dependence range if any included refit exceeds reference-age support, rather than silently dropping its tail.

Input	Female support	Male support	Unit
Systolic BP	88.000 to 153.333	95.000 to 156.248	mmHg
Diastolic BP	47.341 to 95.000	42.667 to 101.333	mmHg
Waist / height	0.402 to 0.847	0.397 to 0.813	ratio
HbA1c	4.400 to 8.600	4.400 to 9.400	%
ApoB	41.000 to 155.000	44.000 to 159.000	mg/dL
hs-CRP	0.100 to 32.713	0.100 to 25.700	mg/L
Cystatin C	0.470 to 1.080	0.542 to 1.148	mg/L
Single-hand grip	20.440 to 44.800	30.289 to 70.049	kg
Estimated VO2max	21.235 to 63.534	29.544 to 80.007	mL/kg/min

The calculator accepts an exercise-based estimate; wearable/non-exercise fitness methods return unsupported for the fitness/full calculation because cross-method calibration was not established. If a laboratory reports a value below a known reporting limit, the entered limit is divided by $\sqrt{2}$ for CRP/cystatin, matching the public-data fill convention. Unknown limits are not assigned invented values. Inputs and drafts are local to the browser; no health values are sent to a server.

9. Frozen equation coefficients

Component	Linear β	Spline θ	Sex slope ψ
Systolic BP	0.0259293641	0.222835824	0.0182375222
Diastolic BP	0.0270364944	0.0489945698	-0.0484816991
Waist / height	-0.0861279657	0.0941047178	-0.154334132
HbA1c	0.275620476	-0.0526873405	0.0302738626
ApoB	-0.0620479836	-0.144321767	-0.107768692
hs-CRP	0.341046175	-0.0288369608	0.0594407397
Cystatin C	0.0982564075	0.134622987	-0.0869846262
Single-hand grip	-0.174573608	-0.0353019203	0.0134526087
Estimated VO2max	-0.106176624	0.0387379731	-0.048278776

The component sex slope multiplies $z(x)(s-0.5)$, not raw x . Coefficients require the exact transformations, knots, centers/scales and baseline. A component taken from another panel cannot be swapped into this full equation while retaining its interpretation: each panel is fitted jointly and calibrated against the reference.

Age-free full-model baseline	Value
intercept	-7.32948245134
male	0.970334272409
gamma_0	0.08779681465
gamma_1	0.0646626040201

Input	Transformed knots	Center / scale
Systolic BP	101.333 / 115.333 / 132.667	116.484 / 13.1484
Diastolic BP	58 / 71.3333 / 85	71.3022 / 11.3087
Waist / height	0.445946 / 0.542036 / 0.679291	0.554361 / 0.0936493
HbA1c	4.8 / 5.2 / 5.7	5.3129 / 0.765764
ApoB (log)	4.12713 / 4.48864 / 4.81218	4.47225 / 0.278599
hs-CRP (log)	-1.20397 / 0.405465 / 2.11626	0.407454 / 1.31926
Cystatin C (log)	-0.525939 / -0.328504 / -0.12897	-0.325157 / 0.182517
Single-hand grip (log)	3.31054 / 3.68872 / 4.04263	3.67641 / 0.285085
Estimated VO2max (log)	3.39047 / 3.67704 / 3.96556	3.67681 / 0.235852

All digits, nonlinear centers/scales, panel coefficients, conditional covariance matrices, failed specifications and numerical fixtures are saved in the repository artifacts. The downloadable browser JSON contains the exact evaluated equation. The report's rounded tables are descriptive; calculations use the full stored precision.

10. Reproduction and provenance

Source: LongevityWorldCup.Research/MortalityAge. Run `prepare.py`, `train.py`, `diagnostics.py`, `ph-diagnostics.py`, `transport.py`, `uncertainty.py`, `export.py`, `report.py` in that order in the pinned environment. Run `test_model.py` and `verify-browser.mjs` for independent numerical and unit parity. Raw downloads, respondent data and intermediate outputs remain in ignored `.artifacts/mortality-age`.

Model version: **mortality-age-0.2-9cbd69d1609e**

Frozen plan SHA-256: 4c6643498cdeab3fbff66da04a2e0ded67cff3e887d8d868fc979a07d4ac304f

Browser model SHA-256: c289c1d135d6b132c84e1736b57aa434a61519cf2c94c673f3a4c7ba55da9dc2

64 original public files are recorded with URL, retrieval UTC, byte size, SHA-256, row count and columns in `artifacts/downloads.json`. Joins are one-to-one on SEQN with duplicate checks. SAS file signatures and mortality-file content are verified. Primary fit completion UTC: 2026-10-09T16:46:07.747018+00:00. Cycle/horizon completion UTC: 2026-10-09T16:43:47.310200+00:00. Validation uncertainty completion UTC: 2026-10-09T16:45:42.419672+00:00. Monte Carlo seed: 91237.

Runtime / package	Version
Python	3.14.7
numpy	2.4.3
scipy	1.17.1
pandas	2.3.3
lifelines	0.30.3
matplotlib	3.11.2
reportlab	4.4.10

Verification includes invariance after changing or deleting personal age in risk design and missing-measurement calculations, unconstrained biomarker-coefficient fitting, likelihood gradients against finite differences, exponential limits, observed-row/quadrature equivalence, right-censored risk sets, historical-reference roundtrips, spline tails and Python/browser fixtures. Every panel, bootstrap and dependence refit is checked for absent age predictors and age-invariant outputs. Website checks cover unit conversions, sex-dependent functions, unsupported inputs, loading recovery, editing, birth-date draft removal, mobile/desktop behavior, PDF bytes and discovery exclusion. Code verification is not scientific validation.

This report distinguishes actual temporal performance from scientific identification. The model has no complete-panel external validation, no causal adjustment or competing-risk interpretation, and no calibrated smartwatch conversion. Medication, smoking, disease, ethnicity and socioeconomic conditions are not separately included in the requested panel. Repeated measurements and longitudinal biological change were not modelled. Those limitations are material, not resolved by producing an age number.

11. Primary sources

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